

GALGOTIAS UNIVERSITY

SCHOOL OF ARTIFICIAL INTELLIGENCE

Building Future-Ready AI Talent

LABORATORY ASSIGNMENT

Course Name	Database Management System	Course Code	E2UC302B
Program	B.Tech. (AIML)	Academic Year	2026–27
Semester / Batch	III	Lab Assignment No.	01
Set	SET A	Date	
Faculty In-Charge	Mr. Arunendra Mani Tripathi	Max. Marks	10
Week No.	4	Duration	90 minutes

Course Outcome(s)	CO3: Apply SQL techniques, including aggregate functions and grouped queries, to analyse and retrieve meaningful information from relational databases.
Skill(s) Addressed	SQL Programming, Aggregate Functions, GROUP BY, HAVING, Data Analysis

1. GENERAL INSTRUCTIONS

- Attempt all questions individually.
- Execute all SQL queries in MySQL Workbench (or the prescribed SQL client).
- Use the dataset/schema given below and retain the same table and column names.
- Show both the SQL query and the corresponding output for every question.
- Save the SQL script as RollNo_Name_DBMS_Lab04_SetX.sql.
- Submit the SQL script and required screenshots through the LMS.
- Plagiarised or copied submissions will be awarded zero.

2. DATASET / TABLE SCHEMA

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(30),  
    Department VARCHAR(20),  
    Semester INT,  
    CGPA DECIMAL(3,2),  
    Attendance DECIMAL(5,2)  
);
```

```
INSERT INTO Student VALUES  
(101, 'Rahul', 'AIML', 4, 8.40, 88.5),  
(102, 'Anjali', 'AIML', 4, 9.10, 92.0),  
(103, 'Amit', 'CSE', 4, 7.20, 71.0),  
(104, 'Priya', 'CSE', 4, 8.10, 80.0),  
(105, 'Neha', 'ECE', 4, 8.80, 90.5),  
(106, 'Rohit', 'ECE', 4, 6.90, 68.0),  
(107, 'Simran', 'AIML', 4, 8.00, 74.5),  
(108, 'Farhan', 'CSE', 4, 9.40, 95.0),
```

```
(109, 'Divya', 'ECE', 4, 7.60, 82.0),
(110, 'Karan', 'AIML', 4, 8.90, 89.0);
```

3. QUESTIONS / TASKS

- Q1.** Display the total number of students using COUNT().
- Q2.** Find the total and average CGPA of all students using SUM() and AVG().
- Q3.** Find the minimum and maximum CGPA using MIN() and MAX().
- Q4.** Display the number of students in each department using GROUP BY.
- Q5.** Display the average CGPA of each department using GROUP BY.
- Q6.** Display only those departments whose average CGPA is greater than 8.0 using HAVING.
- Q7.** Display departments having more than 3 students using GROUP BY and HAVING.
- Q8.** Display the minimum and maximum attendance for each department.

4. STUDY MATERIAL / REFERENCE

- MySQL 8.0 Reference Manual: <https://dev.mysql.com/doc/refman/8.0/en/>
- Aggregate Functions: <https://dev.mysql.com/doc/refman/8.0/en/aggregate-functions.html>
- GROUP BY Optimization / Grouping: <https://dev.mysql.com/doc/refman/8.0/en/group-by-functions.html>
- SELECT Statement: <https://dev.mysql.com/doc/refman/8.0/en/select.html>

5. EVALUATION RUBRIC

Evaluation Criteria	Marks
Correct use of COUNT, SUM, AVG, MIN and MAX	3
Correct GROUP BY queries	2
Correct HAVING conditions	2
Correct output interpretation and verification	1
SQL script quality and formatting	1
Viva / spot-check	1
TOTAL	10

6. EVALUATION PROCESS

- Submission check – SQL script and outputs received within the deadline.
- Execution check – queries are executed successfully and outputs are verified.
- Rubric scoring – marks are awarded against the stated criteria.
- Viva/spot-check – brief questions may be asked to verify individual understanding.

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LABORATORY ASSIGNMENT

Course Name	Database Management System	Course Code	E2UC302B
Program	B.Tech. (AIML)	Academic Year	2026–27
Semester / Batch	III	Lab Assignment No.	02
Set	SET B	Date	
Faculty In-Charge	Mr. Arunendra Mani Tripathi	Max. Marks	10
Week No.	4	Duration	90 minutes

Course Outcome(s)	CO3: Apply SQL techniques, including aggregate functions and grouped queries, to analyse and retrieve meaningful information from relational databases.
Skill(s) Addressed	SQL Programming, Aggregate Functions, GROUP BY, HAVING, Salary Analysis

1. GENERAL INSTRUCTIONS

- Attempt all questions individually.
- Execute all SQL queries in MySQL Workbench (or the prescribed SQL client).
- Use the dataset/schema given below and retain the same table and column names.
- Show both the SQL query and the corresponding output for every question.
- Save the SQL script as RollNo_Name_DBMS_Lab04_SetX.sql.
- Submit the SQL script and required screenshots through the LMS.
- Plagiarised or copied submissions will be awarded zero.

2. DATASET / TABLE SCHEMA

```
CREATE TABLE Employee (
    Emp_ID INT PRIMARY KEY,
    Emp_Name VARCHAR(30),
    Department VARCHAR(20),
    Salary DECIMAL(10,2),
    Experience INT
);
```

```
INSERT INTO Employee VALUES
(1, 'Rahul', 'IT', 55000, 3),
(2, 'Simran', 'IT', 72000, 5),
(3, 'Farhan', 'HR', 48000, 2),
(4, 'Divya', 'Sales', 39000, 2),
(5, 'Vikram', 'IT', 65000, 4),
(6, 'Meera', 'HR', 61000, 6),
(7, 'Aman', 'Sales', 45500, 3),
(8, 'Anjali', 'IT', 83000, 7),
```

```
(9, 'Rohit', 'HR', 52000, 4),
(10, 'Priya', 'Sales', 47000, 5);
```

3. QUESTIONS / TASKS

- Q1.** Display the total number of employees using COUNT().
- Q2.** Find the total and average salary of all employees using SUM() and AVG().
- Q3.** Find the minimum and maximum salary using MIN() and MAX().
- Q4.** Display the number of employees in each department.
- Q5.** Display the average salary of each department.
- Q6.** Display only departments whose average salary is greater than ₹55,000 using HAVING.
- Q7.** Display departments having more than 2 employees.
- Q8.** Display the total salary expenditure for each department.

4. STUDY MATERIAL / REFERENCE

- MySQL 8.0 Reference Manual: <https://dev.mysql.com/doc/refman/8.0/en/>
- Aggregate Functions: <https://dev.mysql.com/doc/refman/8.0/en/aggregate-functions.html>
- GROUP BY Optimization / Grouping: <https://dev.mysql.com/doc/refman/8.0/en/group-by-functions.html>
- SELECT Statement: <https://dev.mysql.com/doc/refman/8.0/en/select.html>

5. EVALUATION RUBRIC

Evaluation Criteria	Marks
Correct use of COUNT, SUM, AVG, MIN and MAX	3
Correct GROUP BY queries	2
Correct HAVING conditions	2
Correct output interpretation and verification	1
SQL script quality and formatting	1
Viva / spot-check	1
TOTAL	10

6. EVALUATION PROCESS

- Submission check – SQL script and outputs received within the deadline.
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Course Name	Database Management System	Course Code	E2UC302B
Program	B.Tech. (AIML)	Academic Year	2026–27
Semester / Batch	III	Lab Assignment No.	03
Set	SET C	Date	
Faculty In-Charge	Mr. Arunendra Mani Tripathi	Max. Marks	10
Week No.	4	Duration	90 minutes

Course Outcome(s)	CO3: Apply SQL techniques, including aggregate functions and grouped queries, to analyse and retrieve meaningful information from relational databases.
Skill(s) Addressed	SQL Programming, Aggregate Functions, GROUP BY, HAVING, Healthcare Data Analysis

1. GENERAL INSTRUCTIONS

- Attempt all questions individually.
- Execute all SQL queries in MySQL Workbench (or the prescribed SQL client).
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2. DATASET / TABLE SCHEMA

```
CREATE TABLE Patient (
    Patient_ID INT PRIMARY KEY,
    Patient_Name VARCHAR(30),
    Department VARCHAR(20),
    Age INT,
    Bill_Amount DECIMAL(10,2)
);
```

```
INSERT INTO Patient VALUES
(1, 'Rehan', 'Cardiology', 54, 85000),
(2, 'Zoya', 'Pulmonology', 29, 42000),
(3, 'Ishaan', 'Cardiology', 62, 95000),
(4, 'Tanya', 'General', 45, 25000),
(5, 'Aryan', 'Cardiology', 70, 110000),
(6, 'Kavya', 'Pulmonology', 38, 38000),
(7, 'Suresh', 'General', 58, 31000),
(8, 'Meera', 'Pulmonology', 33, 47000),
```

```
(9, 'Kabir', 'Cardiology', 67, 102000),
(10, 'Riya', 'General', 49, 28000);
```

3. QUESTIONS / TASKS

- Q1.** Display the total number of patients using COUNT().
- Q2.** Find the total and average bill amount using SUM() and AVG().
- Q3.** Find the youngest and oldest patient using MIN() and MAX() on Age.
- Q4.** Display the number of patients in each department.
- Q5.** Display the average bill amount for each department.
- Q6.** Display only departments whose average bill amount is greater than ₹40,000 using HAVING.
- Q7.** Display departments having at least 3 patients.
- Q8.** Display the total bill amount collected from each department.

4. STUDY MATERIAL / REFERENCE

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- Aggregate Functions: <https://dev.mysql.com/doc/refman/8.0/en/aggregate-functions.html>
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